INCH-POUND

NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

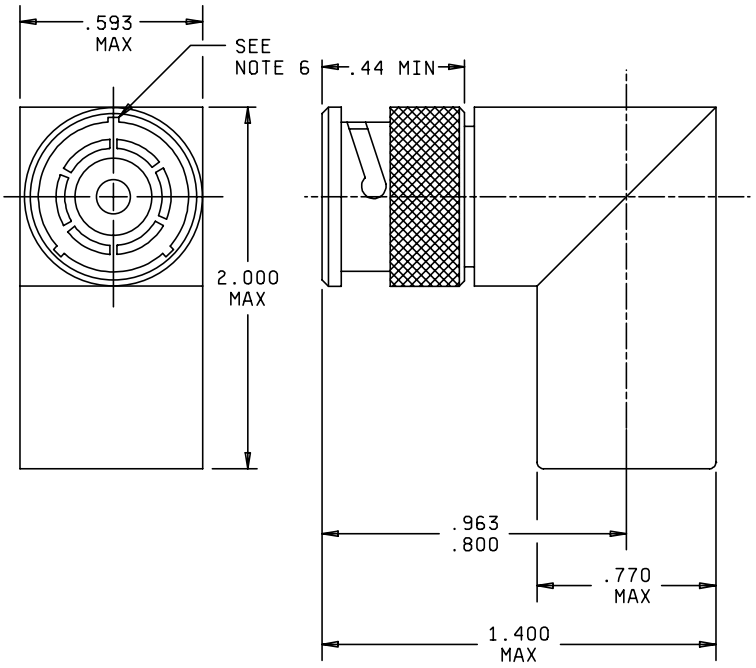
MIL-PRF-49142/7B
5 November 1990
SUPERSEDING
MIL-C-49142/7A
13 September 1989

PERFORMANCE SPECIFICATION

CONNECTOR, PLUG, ELECTRICAL, TRIAXIAL, RADIO FREQUENCY
(SERIES TRB (CABLED), PIN CONTACT, RIGHT ANGLE, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-49142



Inches	mm
.44	11.18
.510	12.95
.593	15.06
.770	19.56
.800	20.32
.963	24.46
1.400	35.56
2.000	50.80

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. 2.000 (50.80 mm) defines the maximum length of the connector when assembled to the appropriate cable.
4. Wrench flats are to accommodate standard wrench openings in accordance with FED-STD-H28.
5. All undimensioned pictorial representations are for reference purposes only.
6. Alternate keying configurations, see MIL-PRF-49142, figure 1.
7. Interface shall be in accordance with MIL-STD-348, series TRB, pin contact.

FIGURE 1. General configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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ENGINEERING DATA:

Nominal impedance: Nonconstant.

Frequency range: 0 to 500 MHz minimum.

Voltage rating: 400 V rms maximum working voltage at sea level. 100 V rms maximum working voltage at 70,000 feet.

Temperature range: -65 to +165 degrees Celsius.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and MIL-STD-348.

Force to engage and disengage:

Longitudinal force: 4 pounds maximum.

Torque: 2.5 inch-pounds maximum.

Coupling proof torque: Not applicable.

Mating characteristics: See MIL-STD-348 for dimensions.

Intermediate contact:

Test ring ID: .169 inch maximum, 16-microinch finish.

Insertion force: 3 pounds maximum, when inserted a minimum of .093 inch.

Contacts with slotted members: Shall contact a .173 inch minimum diameter ring within .031 inch of their tip ends.

Outer contact:

Test ring ID: .319 inch maximum, 16-microinch finish.

Insertion force: 5 pounds maximum, when inserted a minimum of .093 inch.

Contacts with slotted members: Shall contact a .324 inch diameter ring within .031 inch of their tip ends.

Permeability: Applicable.

Hermetic seal: Not applicable.

Leakage: To be added.

Insulation resistance: 5,000 megohms.

Outer conductor retention: 6 pounds minimum axial force.

Dielectric withstanding voltage: At sea level, 1,200 V rms, between center conductor and intermediate conductor; 500 V rms, between intermediate conductor and outer conductor.

Salt spray (corrosion): Applicable.

Vibration: Applicable.

Shock: Applicable.

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Thermal shock: Applicable (except high test temperature shall be 200 degrees Celsius for connectors using 200 degree Celsius cables).

Moisture resistance: Applicable.

Conductor resistance: In milliohms maximum.

	Initial	After environment
Center conductor	2.0	2.5
Intermediate conductor:	0.5	0.6
Outer conductor:	0.2	0.3

TABLE I. Dash number and applicable cable.

Dash number *	Cable 1/
Category A - No special tools required 2/ 3/	
0001 4/ 0101 0201	M17/134-00001
0002 4/ 0102 0202	M17/134-00002
0003 5/	D3-7619-5/336
0004 5/	D3-7619-5/338
0005 5/	M17/116-RG307
0006 4/ 6/, 3006 4/ 6/ 0106 6/, 3106 6/ 0206 6/, 3206 6/	M17/176-00002 7/
0007 4/ 0107 0207	M17/177-00001 7/
0008 4/ 0108 0208	M17/178-00001
0009 4/ 0109 0209	M17/179-00001

See footnotes at end of table.

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TABLE I. Dash number and applicable cable - Continued.

Dash number *	Cable 1/
Category G - Use of MIL-C-22520 tool required for assembly 2/ 8/ 9/	
0013 4/ 0113 0213	M17/134-00001
0014 4/ 0114 0214	M17/134-00002
0015 4/ 0115 0215	M17/116-RG307
0016 4/ 0116 0216 3016 4/ 3116 3216	M17/45-RG108
0017 4/ 0117 0217 3017 4/ 3117 3217	M17/176-00002 7/
0018 4/ 0118 0218	M17/177-00001 7/
0019 4/ 0119 0219	M17/178-00001
0020 4/ 0120 0220	M17/179-00001

* Not for Naval Air Systems Command (AS) use.
1/ The latest version of each cable shall be applicable.
2/ These connectors have captivated center contacts.
3/ Not for Navy use.
4/ Preferred keying arrangement.
5/ Inactive for new design (see table III).
6/ Not for Air Force use.
7/ Cables to be used for the 200 degree Celsius thermal shock test.
8/ These connectors are assembled using the applicable crimping tool to the specified cables.
9/ Complete connector assembly shall consist of a body, center contact, intermediate contact, ferrule, and assembly instructions.

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Corona level:

Altitude: 70,000 feet.

Voltage: 250 V rms minimum.

RF high potential withstanding voltage:

800 V rms, between center conductor and intermediate conductor.

200 V rms, between intermediate conductor and outer conductor at 5 MHz to 7.5 MHz.

Leakage current: Not applicable.

Cable retention force (for cable .200 inch to .325 inch outside dimension): 40 pounds minimum.

Coupling mechanism retention force: 100 pounds minimum.

Rise time degradation: 400 picoseconds maximum. 1/

Connector durability: 500 cycles minimum at 12 cycles per minute maximum.

Part or Identifying Number (PIN): M49142/07- (dash number from table I).

TABLE II. Group qualification. 1/

Group	Submission and qualification of any of the following dash numbers	Qualifies the following dash nos.
I	+X01 +X02 +X07	+X01 +X02 +X07
II	+X03 +X04 +X08 +X09	+X03 +X04 +X08 +X09
III	+X06	+X06
IV	+X13 +X14 +X15 +X18	+X13 +X14 +X15 +X18
V	+X16 +X17	+X16 +X17
VI	+X19 +X20	+X19 +X20

1/ If a connector manufacturer produces a connector which meets all the requirements for two or more connector PINs (within same series), the manufacturer may receive qualification approval for two or more connector PINs by qualifying the one connector. Each connector, however, must be marked with its own appropriate PIN. For group qualification, the connectors must be of similar design. Qualification of connectors qualifies connectors of the same body material and finish only. + designates body material and finish. X designates keying (see MIL-C-49142).

1/ Not applicable to connectors using twin conductor cables.

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TABLE III. Cross-reference of cables.

Preferred cable	Superseded cable
M17/177-00001	350-10045-1
M17/178-00001	D3-7619-5/336
M17/179-00001	D3-7619-5/338

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 99

User activities:
Army - AT
Navy - AS, MC, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3798-05)

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